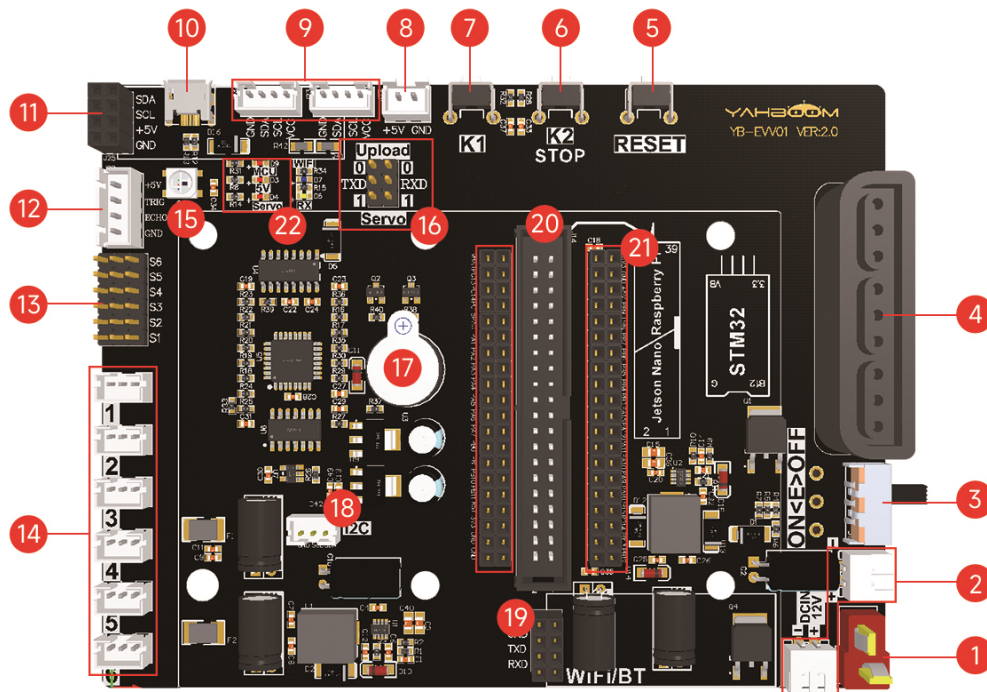


Expansion Board Introduction

1. Expansion Board Introduction

The Dofbot-SE expansion board is compatible with multiple main controllers, but only one main controller can be connected to the expansion board at a time. It supports Raspberry Pi, Jetson Nano, STM32, and also has a reserved external I2C interface that can support other main controller boards with I2C functionality.

2. Component Locations and Functions on the Expansion Board



- ① T-type power supply interface: 12V power adapter to power supply robotic arm expansion board.
- ② 12V output interface: Provides 2PCS XH2.0 interfaces that output 12V.
- ③ Power switch: Robotic arm power switch.
- ④ PS2 controller receiver base: Install PS2 controller receiver.
- ⑤ RESET key: Reset the MCU and STM32.
- ⑥ K2: Short press: the bus servo stops suddenly and turns off the torque of the bus servo; long press for about 10 seconds: closes and opens the BootLoader function of the underlying microcontroller. After the BootLoader function is opened, the RGB light shows a marquee effect.
- ⑦ K1: In default mode, press the K1 key once to reset the bus servo and raise the robotic arm vertically. Double-click the K1 key to quickly reset the servo.
- ⑧ 5V output interface.
- ⑨ I2C interface: Can be used to connect external I2C devices.
- ⑩ microUSB interface: Write programs for STM32, update firmware for MCU and communicate with the main control board through the serial port.
- ⑪ I2C interface: Connect OLED screen.
- ⑫ Ultrasonic interface: Connect ultrasonic module.
- ⑬ PWM servo interface: Connect PWM servo.
- ⑭ Bus servo interface: Connect the bus servo.
- ⑮ RGB lights: RGB lights controlled by the underlying microcontroller. The main control board B can send commands to the microcontroller to make the RGB lights light up in different colors.
- ⑯ MCU serial port function selection: In the default state, use a jumper cap to connect both TXD and RXD to 1 (Servo). Only the MCU can control the bus servo. If you use a jumper cap to connect both TXD and RXD to 0 (Upload), you can use microUSB interface to update the firmware of MCU.
- ⑰ Buzzer: Active buzzer.
- ⑱ I2C interface: Communication interface between Raspberry Pi 5 and robotic arm expansion board.
- ⑲ Serial port interface: Connect WiFi module or Bluetooth module.
- ⑳ Jetson Nano and robotic arm expansion board connection interface .
- ㉑ STM32 core board and robotic arm expansion board connection interface.
- ㉒ Status indicator light:

MCU: The status indicator light of MCU. When the microcontroller is running normally, the red light flashes twice every 3 seconds.

5V: The 5V voltage indicator light of the expansion board. When the 5V power supply is normal, it is always on;

Servo: The servo power supply indicator light. When the servo is powered normally, it keep on.

Special Note: The 22.04 version VM comes with a large model voice interaction module, replacing the voice recognition and voice broadcast module on I2C interface #9. If you purchased the module previously and want to experience ROS2 related cases, the voice-related cases are not universally applicable.